SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Computer Networks **COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO5: *Measure network performance using key metrics with simulation tools*

Assessment Tool Weightage: 20%

QUESTIONS: 40 **Total Marks:40**

Annexure A

Mock Interview

QUESTIONS:40 **Total Marks:40**

1. Analyze how network congestion affects packet delay and packet loss. (BL4)(1)
2. Compare the impact of star and mesh topologies on network reliability and fault tolerance. (BL4)(1)
3. Why does increasing the number of routers between source and destination increase end-to-end delay? Explain logically. (BL4)(1)
4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic. (BL4)(1)
5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication. (BL5)(1)
6. Analyze how network segmentation using VLANs improves performance and security. (BL4)(1)
7. Compare the effects of TCP and UDP on network performance for different applications. (BL4)(1)
8. Why is load balancing important in large enterprise networks? Analyze its impact on performance and availability. (BL5)(1)
9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization. (BL4)(1)
10. Evaluate the trade-off between network security measures (such as firewalls and encryption) and overall network performance. (BL5)(1)
11. A network has high delay—how will you identify and fix the issue? (BL4)(1)

12. If throughput is low despite high bandwidth, what could be the reasons? (BL4)(1)
13. How would you reduce packet loss in a congested network? (BL4)(1)
14. A VoIP call is breaking frequently—how would you troubleshoot? (BL4)(1)
15. Given a network with high jitter, suggest improvements (BL4)(1)
16. How will you simulate a congested network in Packet Tracer? (BL5)(1)
17. If packets are delayed in a router, what steps will you take? (BL4)(1)
18. How would you optimize a network for online gaming? (BL4)(1)
19. Design a small network and explain how you will measure its performance (BL5)(1)
20. If RTT is very high, how will you diagnose the issue? (BL4)(1)
21. Explain network performance to a non-technical person (BL5)(1)
22. Describe throughput using a real-life example (BL5)(1)
23. Present how packet loss affects video streaming (BL5)(1)
24. Explain delay types clearly in under 1 minute (BL4)(1)
25. Describe Cisco Packet Tracer in a professional way (BL5)(1)
26. Explain QoS in simple and technical terms (BL5)(1)
27. How would you present your simulation results to a client? (BL4)(1)
28. Explain jitter using a real-world analogy (BL5)(1)
29. Describe how you tested network performance in a lab (BL4)(1)
30. Explain the importance of performance metrics in one structured answer (BL5)(1)
31. Introduce yourself and explain your knowledge of network performance (BL5)(1)
32. Why are you interested in networking and simulation tools? (BL5)(1)
33. What makes you confident in analyzing network performance? (BL5)(1)
34. Explain a concept you are very strong in (related to networking) (BL5)(1)
35. How would you handle a question you don't know? (BL5)(1)
36. Describe a situation where you solved a technical problem (BL4)(1)
37. Why should we select you for a networking role? (BL5)(1)
38. What challenges did you face while learning Packet Tracer? (BL5)(1)
39. How do you improve your technical skills regularly? (BL5)(1)
40. Demonstrate confidence while explaining a complex concept simply (BL5)(1)

1. Network congestion and packet delay/loss

Network congestion occurs when the amount of traffic exceeds the network's available capacity. Routers then queue packets, increasing delay and jitter. When buffers become full, packets are dropped, causing packet loss and retransmissions, which further reduce performance.

2. Star vs Mesh: reliability and fault tolerance

Star topology has good fault isolation because failure of one link affects only one device, but failure of the central switch can affect the entire network. Mesh topology has multiple paths between nodes, so traffic can be rerouted when a link fails. Therefore, mesh provides higher fault tolerance but is more expensive and complex.

3. More routers and end-to-end delay

Every router introduces processing, queuing, and transmission delays. Packets also spend time propagating over each link. Therefore, as the number of routers increases, the total end-to-end delay generally increases.

4. QoS for voice and video

QoS prioritizes delay-sensitive traffic such as VoIP and video over less-critical traffic. It can provide priority queuing, bandwidth reservation, congestion management, and traffic classification. This reduces delay, jitter, and packet loss, improving call and video quality.

5. Fiber optic vs copper

Fiber provides higher bandwidth, longer transmission distances, lower attenuation, and immunity to electromagnetic interference compared with copper. It is excellent for high-speed backbone communication. However, fiber equipment and installation can initially cost more.

6. VLANs and performance/security

VLANs divide a physical network into logical broadcast domains. This reduces unnecessary broadcast traffic and can improve performance. VLANs also isolate

groups such as students, employees, and guests, improving security when combined with ACLs and firewall rules.

7. TCP vs UDP

TCP provides reliable, ordered delivery using acknowledgments, retransmissions, and congestion control, but these mechanisms add overhead and latency. UDP has lower overhead and no built-in delivery guarantee, making it suitable for applications such as live video, VoIP, and gaming where speed is often more important than perfect delivery.

8. Importance of load balancing

Load balancing distributes traffic across multiple servers or network paths. It prevents one resource from becoming overloaded, improving response time, scalability, and availability. If one server fails, traffic can be redirected to another available server.

9. Frequent routing updates

Routing updates help routers maintain accurate paths, but excessive updates consume bandwidth, CPU, and memory. They can also temporarily increase processing overhead. Efficient routing protocols use mechanisms such as triggered updates and route summarization to reduce unnecessary traffic.

10. Security vs performance trade-off

Firewalls, encryption, IDS/IPS, and authentication improve security but require processing resources. Strong encryption and deep packet inspection can increase latency and reduce throughput if poorly designed. The solution is to use appropriately sized hardware, optimized policies, and selective inspection while maintaining required security.

11. Network has high delay—how to identify and fix it

First, use ping and traceroute to locate where delay occurs. Check router CPU usage, interface utilization, queue sizes, and link errors. Then identify congestion, faulty links, inefficient routing, or overloaded devices and fix them using QoS, routing optimization, bandwidth upgrades, or hardware replacement.

12. Low throughput despite high bandwidth

Possible causes include:

- ***High latency***
- ***Packet loss***
- ***Network congestion***
- ***Poor Wi-Fi signal***
- ***Faulty cables/interfaces***
- ***Server limitations***
- ***TCP congestion control***
- ***Router/firewall bottlenecks***

Thus, bandwidth represents capacity, while actual throughput depends on several other factors.

13. Reducing packet loss in congestion

Use QoS and traffic prioritization, increase available bandwidth, remove bottlenecks, optimize routing, and monitor overloaded interfaces. For wireless networks, improve AP placement and reduce interference. Packet loss should be measured before and after changes to verify improvement.

14. VoIP call frequently breaking

I would check:

1. ***Packet loss***
2. ***Jitter***
3. ***RTT/latency***
4. ***Bandwidth utilization***
5. ***QoS configuration***
6. ***Wi-Fi signal quality***
7. ***VoIP device and SIP configuration***

Then prioritize voice traffic with QoS and investigate any congested or faulty network segment.

15. Improving high jitter

Identify the source using network monitoring tools, then reduce congestion, configure QoS, prioritize real-time traffic, optimize routing, and improve unstable wireless links. A stable and appropriately sized network reduces variation in packet arrival times.

16. Simulating congestion in Packet Tracer

In Cisco Packet Tracer, I would create multiple hosts and routers, generate traffic between multiple endpoints, and configure a relatively limited-bandwidth link. I would then monitor traffic using simulation mode and observe packet queues, delays, drops, and retransmissions. Different bandwidth and traffic conditions can be compared.

17. Packets delayed in a router

I would:

1. ***Check interface utilization.***
2. ***Check router CPU and memory.***
3. ***Examine queue and buffer conditions.***
4. ***Check interface errors.***
5. ***Use ping/traceroute to locate the delay.***
6. ***Check routing paths.***
7. ***Apply QoS or upgrade the bottleneck if required.***

18. Optimizing a network for online gaming

Gaming requires low latency, low jitter, and minimal packet loss. I would use wired Ethernet where possible, prioritize gaming traffic with QoS, avoid congested Wi-Fi channels, use efficient routing, and prevent background downloads from consuming bandwidth.

19. Small network and performance measurement

PC1 ── Switch ── Router ── Server
PC2 ──┘

I would measure:

- ***Bandwidth: link capacity***

- ***Throughput: actual data delivered***
- ***Latency/RTT: ping***
- ***Packet loss: percentage of lost packets***
- ***Jitter: variation in packet delay***

I would record baseline measurements, introduce traffic, measure again, and compare the results.

20. Diagnosing high RTT

Use `ping` to determine whether RTT is consistently high. Then use `traceroute` to identify the network segment introducing delay. Check routing, congestion, overloaded devices, physical links, and geographic distance. Compare results from multiple endpoints to determine whether the problem is local or remote.

Presentation / Explanation Questions

21. Network performance to a non-technical person

Network performance is like the quality of a road. Bandwidth is the number of lanes, throughput is how many vehicles actually reach their destination, delay is how long each vehicle takes, and packet loss is like vehicles disappearing before reaching the destination.

22. Throughput real-life example

Imagine a water pipe that can theoretically carry 100 liters per minute. If only 70 liters actually reach the destination each minute, the actual flow is 70 liters/minute. Similarly, network throughput is the actual amount of data successfully delivered per unit time.

23. Packet loss and video streaming

Packet loss means some video data does not reach the viewer. This can cause freezing, buffering, pixelation, missing frames, or reduced video quality. Small

losses may be hidden by buffering or error recovery, but high loss significantly affects the viewing experience.

24. Types of delay in under one minute

There are four major delays: processing delay for examining a packet, queuing delay while waiting in a router, transmission delay for putting the packet onto the link, and propagation delay for the signal to travel through the medium. Together they contribute to end-to-end delay.

25. Cisco Packet Tracer professionally

Cisco Packet Tracer is a network simulation and learning platform that allows users to design, configure, and troubleshoot virtual networks. It supports devices such as routers and switches and provides simulation capabilities for analyzing network behavior without requiring physical hardware.

26. QoS in simple and technical terms

Simple: QoS is like giving an ambulance priority on a busy road.

Technical: Quality of Service classifies network traffic and assigns appropriate treatment using mechanisms such as classification, marking, queuing, scheduling, policing, and shaping. Real-time traffic can therefore receive priority during congestion.

27. Presenting simulation results to a client

I would present:

1. *Objective*
2. *Network topology*
3. *Test conditions*
4. *Baseline metrics*
5. *Results using tables/graphs*
6. *Problems identified*
7. *Improvements implemented*
8. *Before-and-after comparison*
9. *Final recommendation*

I would focus on business impact rather than only technical terminology.

28. Jitter real-world analogy

Imagine a bus that should arrive every 10 minutes. If it arrives exactly every 10 minutes, the schedule is stable. If it arrives after 5 minutes, then 20 minutes, then 7 minutes, the variation is similar to network jitter—packets are arriving with inconsistent delays.

29. Testing network performance in a lab

I would first build the topology and establish connectivity. Then I would generate normal and heavy traffic and measure throughput, latency, packet loss, and jitter using tools such as ping, traceroute, Packet Tracer simulation, or network monitoring software. Finally, I would compare the measurements and identify bottlenecks.

30. Importance of performance metrics

Network performance metrics provide measurable information about network health.

Metric	What it tells us
Bandwidth	Maximum capacity
Throughput	Actual delivered data
Latency	Communication delay
Packet loss	Packets that fail to arrive
Jitter	Variation in delay

Together, these metrics help identify bottlenecks, evaluate applications, troubleshoot problems, and verify whether network improvements are successful.

Interview / Self-Introduction Questions

For these questions, you can answer in the first person during a viva or interview.

31. Introduce yourself and explain your networking knowledge

I am a Computer Science student with an interest in computer networks and network performance. I have learned concepts such as topologies, IP addressing, routing, switching, TCP/IP, QoS, latency, throughput, packet loss, and jitter. I have also practiced designing and analyzing networks using Cisco Packet Tracer.

32. Why are you interested in networking and simulation tools?

I am interested in networking because modern applications depend on reliable communication between devices. Simulation tools such as Packet Tracer allow me to understand routing, switching, troubleshooting, and performance without requiring physical hardware. They also help me experiment safely with different network configurations.

33. What makes you confident in analyzing network performance?

I understand the major performance metrics and how they are related. I can use tools such as ping, traceroute, and network simulation to identify latency, packet loss, congestion, and routing problems. I also prefer to verify a problem using measurements rather than making assumptions.

34. Explain a concept you are very strong in

One concept I am particularly comfortable with is QoS. QoS classifies traffic and gives important applications appropriate priority. For example, voice traffic requires low delay and jitter, so it can be prioritized over less time-sensitive traffic such as large file downloads.

35. How would you handle a question you don't know?

I would be honest that I do not know the answer instead of guessing. I would explain what I understand about the related concept, then investigate the correct information using reliable documentation or testing. I see unknown questions as opportunities to learn.

36. Describe a situation where you solved a technical problem

During a network lab, if two devices could not communicate, I would troubleshoot systematically. I would first check the physical connections, then IP addresses and subnet masks, followed by interface status, routing, and firewall or ACL rules. By testing each layer step by step, I can isolate the cause instead of changing multiple settings at once.

37. Why should we select you for a networking role?

I have a strong interest in networking and a willingness to learn. I understand fundamental concepts such as routing, switching, network performance, security, and troubleshooting. As a fresher, I may not know every advanced technology yet, but I am comfortable learning, testing, analyzing problems logically, and improving my skills continuously.

38. Challenges while learning Packet Tracer

Initially, understanding how routers, switches, interfaces, IP addresses, and routing protocols work together was challenging. I improved by building small networks first and gradually increasing their complexity. Repeated configuration and troubleshooting helped me understand the concepts practically.

39. How do you improve technical skills regularly?

I practice programming and networking concepts regularly, build small projects and simulations, review documentation, and troubleshoot different scenarios. I also try to understand why a solution works rather than simply memorizing commands.

40. Explain a complex concept simply

Consider routing. Imagine sending a parcel from Chennai to Delhi. The parcel may pass through several cities before reaching its destination. Similarly, a router examines the destination IP address and chooses an appropriate next hop. Multiple routers work together to move the packet from its source network to its destination.

Strong closing line for a viva:

> ***"I try to understand networking through both theory and practical simulation. I use performance metrics such as latency, throughput, packet loss, and jitter to measure the actual behavior of a network and then use that information to identify and solve problems."***

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions